
DELIVERY-AWARE PHARMACOPHORE-GUIDED 3D DIFFUSION FOR CNS DE NOVO MOLECULE DESIGN WITH BBB PENETRATION AND LNP DELIVERABILITY

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ABSTRACT

We address the formidable challenge of designing CNS-active molecules that simultaneously achieve blood–brain barrier permeation, robust target engagement, and favorable lipid nanoparticle formulation while remaining synthetically tractable, a problem made difficult by data sparsity and competing objectives. We propose NeuroPharmDiff-MultiBio, a pharmacophore-conditioned SE(3)-equivariant diffusion framework augmented with differentiable multi-task CNS predictors, a differentiable retrosynthesis scorer, and an active-learning loop to efficiently acquire CNS-relevant labels. The method represents molecules as 3D graphs and uses a pharmacophore encoder to produce a continuous conditioning vector that guides a geometry-aware diffusion generator, which is further informed by delivery and feasibility cues. A suite of differentiable predictors maps generated molecules to CNS activity, BBB permeability, LNP encapsulation propensity, metabolic stability, and toxicity, all producing calibrated uncertainties that steer gradient-based guidance during generation and acquisition decisions. A differentiable retrosynthesis module rewards synthetically feasible scaffolds, enabling cost-efficient translation from in silico proposals to real-world routes. We adopt a hybrid multi-objective optimization strategy that combines uncertainty-aware scalarization with Pareto-front selection to balance CNS efficacy, delivery, and synthetic feasibility. Our experimental plan comprises three phases: (i) baseline and core-component validation against diffusion baselines and pharmacophore-guided heuristics, (ii) integrated training with ablations and Pareto analyses, and (iii) an active-learning cycle pairing generated candidates with in silico/in vitro labels and synthesis planning, progressing toward top CNS candidates with scalable synthesis and early permeability/target-binding validation. The proposed work delivers a unified, cost-efficient CNS design pipeline with explicit consideration of BBB penetration, LNP deliverability, and practical synthesis.

1 Introduction

Designing CNS-active molecules that cross the blood–brain barrier (BBB), engage CNS targets, and are amenable to lipid nanoparticle (LNP) formulation remains a formidable challenge in drug discovery. Recent work demonstrates that knowledge-guided diffusion can map pharmacophore features in 3D space to guide design [1]. Pharmacophore-conditioned diffusion approaches for ligand-based 3D design show how domain-informed constraints can steer generation toward desired interaction patterns [2]. This is complemented by advances in structure-aware diffusion: SE(3)-equivariant diffusion enables conditioning on three-dimensional protein environments, enabling geometry-aware design [3]. This approach is complemented by SE(3)-equivariant diffusion methods for structure-aware design in similar contexts. However, these strands have not explicitly integrated CNS-specific properties such as BBB penetrance or formulation constraints.

This gap matters because CNS attrition stems from poor brain exposure, insufficient target engagement, and formulation-related hurdles. Prior work demonstrates that embedding pharmacophore and geometry constraints improves alignment with bioactivity and docking objectives [2]. This alignment is further supported by pharmacophore-guided diffusion approaches [1]. Yet the CNS-specific conditioning, including BBB permeation and LNP deliverability, remains underexplored.

The difficulty is compounded by data sparsity for CNS-relevant properties and the need for differentiable multi-task predictors that generalize to novel chemistries. Synthesizability and retrosynthetic feasibility are essential to convert generated candidates into deliverable molecules, but integrating these with CNS-focused objectives is nontrivial [4] [5].

Previous efforts have advanced modular components (pharmacophore-conditioned generation, SE(3) diffusion, retrosynthesis planning) but have not combined them into a coherent CNS-focused pipeline [2, 3]. This motivates a unified framework that jointly optimizes BBB/CNS activity, LNP compatibility, and synthetic feasibility.

We propose NeuroPharmDiff-MultiBio, a pharmacophore-conditioned SE(3)-equivariant diffusion framework augmented with differentiable, multi-task CNS predictors and a differentiable retrosynthesis module. The approach comprises: (i) a 3D graph representation and pharmacophore encoder; (ii) a multi-objective diffusion generator; (iii) active learning for CNS labels; (iv) a synthetic-feasibility scorer. We anticipate improved CNS-target engagement, BBB-penetration consistency, and scalable, cost-efficient synthetic routes for CNS de novo design.

2 Related Work

Pharmacophore-conditioned diffusion for ligand-based 3D design Pharmacophore-driven diffusion approaches demonstrate how abstract pharmacophore fingerprints can steer 3D molecular generation toward bioactive shapes. In particular, pharmacophore-conditioned diffusion models for ligand-based de novo drug design show that integrating pharmacophore hypotheses into a generative process yields molecules that better satisfy targeted interaction features and docking objectives, even without explicit protein structures. This line of work establishes the viability of conditioning generative models on domain-informed constraints to achieve higher alignment with desired bioactivity patterns. However, these methods have not (yet) explicitly integrated CNS-specific properties such as BBB penetrance or CNS-targeted formulation considerations, which are central to our CNS-focused objective. [2]

SE(3)-equivariant diffusion for structure-based drug design Recent advances recast structure-based drug design as a 3D conditional generation problem using SE(3)-equivariant diffusion. Diffusion models like DiffSBDD generate novel ligands conditioned on protein pockets, enabling explicit incorporation of three-dimensional structural information and enabling constraints such as pocket geometry and binding considerations. This work demonstrates that a single 3D diffusion framework can support diverse objectives (e.g., docking compatibility, targeted design, and inpainting) when conditioned on protein environments, underscoring the potential of geometry-aware generative design for CNS-targeted ligands where binding site context is crucial. These insights motivate integrating pocket- and geometry-conditioned diffusion with additional CNS-relevant conditioning in our proposal. [3]

Synthesizability and retrosynthetic planning in de novo design A central bottleneck in practical de novo design is ensuring synthetic feasibility. Work introducing data-driven evaluators of molecule synthesizability leverages retrosynthetic planning and reaction prediction to assess whether proposed molecules can be realistically constructed, moving beyond simple synthetic accessibility scores. This perspective is essential for cost-efficient pipelines, as it couples generation with plausible synthetic routes and guides optimization toward practically deliverable candidates. Such retrosynthesis-informed evaluation complements pharmacophore and structural conditioning by narrowing the design space to synthetically tractable chemistries. [4]

3 Method

Problem definition We formalize CNS-aware de novo molecule design as a conditional generation problem. Let $\mathcal{P}$ denote the space of pharmacophore hypotheses (abstract feature sets specifying hydrogen-bond donors/acceptors, aromatic centers, cationic/anionic loci, spatial constraints), let $\mathcal{G}$ denote optional protein pocket geometry or binding-site context, and let $\mathcal{Y}$ denote the space of chemically valid 3D molecular graphs:

$$y = (V, E, X, A) \in \mathcal{Y}, \quad (1)$$

where V is the set of atoms, E the bond set, $X = \{x_i\}_{i=1}^{|V|}$ with $x_i \in \mathbb{R}^3$ the Cartesian coordinates, and $A = \{a_i\}_{i=1}^{|V|}$ the atomic types. The design task is to learn a conditional generator

$$\mathcal{M}_\theta : \mathcal{P} \times \mathcal{G} \times \mathcal{C} \rightarrow \mathcal{Y}, \quad (2)$$

that maps a pharmacophore hypothesis $p \in \mathcal{P}$, optional pocket geometry $g \in \mathcal{G}$, and auxiliary conditioning information $c \in \mathcal{C}$ (delivery and synthetic-feasibility cues) to molecules y . The auxiliary conditioning c is produced by differentiable predictors described below and encodes BBB permeability propensity, LNP-encapsulation likelihood, and synthetic-accessibility preference. Our objective is to produce samples from the target conditional distribution $p_\theta(y \mid p, g, c)$ that simultaneously satisfy geometric pharmacophore constraints, predicted CNS-relevant properties, and synthetic deliverability.

Representation and pharmacophore encoder: what and why Representing molecules as 3D graphs with explicit coordinates preserves geometry necessary for CNS target engagement and for evaluating both docking and LNP compatibility; this motivates a geometry-aware generation backbone. Pharmacophore hypotheses p are encoded into a continuous conditioning vector $h_p \in \mathbb{R}^{d_p}$ via a differentiable encoder $\mathcal{E}_\psi : \mathcal{P} \rightarrow \mathbb{R}^{d_p}$. Formally,

$$h_p = \mathcal{E}_\psi(p), \quad (3)$$

where $\mathcal{E}_\psi$ aggregates (i) pharmacophore feature types, (ii) relative spatial constraints (pairwise distances/angles), and (iii) optional residue-contact priors from g . The encoder is designed to be permutation-invariant over pharmacophore points and to output features at multiple spatial scales so that conditioning can modulate both local functional-group placement and global scaffold geometry.

Rationale: a continuous, differentiable pharmacophore representation allows direct gradient-based guidance of the generative model and facilitates soft constraint satisfaction (favoring but not deterministically enforcing exact placements), which is necessary to trade off pharmacophore fidelity against delivery and synthesizability objectives.

SE(3)-equivariant diffusion generator: formulation and rationale We build the generative backbone as a continuous-time diffusion process in the space of 3D molecular graphs, with an SE(3)-equivariant score model $s_\theta(y_t, t \mid h_p, h_c, g)$ that predicts the gradient of the log-density of noisy molecular states y_t conditioned on pharmacophore encoding h_p , auxiliary conditioning h_c (described below), and pocket geometry g . The forward (noising) SDE is:

$$dy_t = f(t)y_t dt + g(t)dw_t, \quad (4)$$

and the reverse-time generative SDE employs s_θ to denoise. In practice we discretize time and operate on node features, coordinates, and bond logits jointly; the score estimator is built from SE(3)-equivariant message-passing blocks to ensure correct transformation behavior under rotations and translations. This follows the design principles of SE(3)-equivariant diffusion for structure-aware molecular generation [3].

Rationale: SE(3)-equivariance enforces geometric consistency and significantly reduces the sample complexity for learning shape-conditioned distributions, which is critical when pharmacophore spatial constraints and binding-site geometries must be respected.

Differentiable multi-task property predictors: roles and integration We define a suite of differentiable predictors $\{f_k\}_{k=1}^K$ that map molecules y to property scores and associated uncertainties:

$$\hat{s}_k, \hat{\sigma}_k = f_k^{\phi_k}(y), \quad k \in \{\text{CNS activity, BBB, LNP, metabolic, toxicity, } \dots\}. \quad (5)$$

Each predictor outputs a differentiable log-likelihood term $\log p_{\phi_k}(s_k \mid y)$ or a surrogate reward $r_k(y)$ so that we can compute gradients $\nabla_y \log p_{\phi_k}(s_k \mid y)$ or $\nabla_y r_k(y)$ for guidance during generation. Predictors are trained as multi-task networks with calibrated uncertainty estimates to support uncertainty-weighted guidance and acquisition in active learning.

Rationale: Treating property estimation as differentiable enables classifier- or score-based guidance of the diffusion sampler via gradient nudging, allowing continuous tradeoffs between geometric fidelity and property optimization without requiring discrete post-filtering.

Guided reverse diffusion and differentiable retrosynthesis scoring To bias generation toward desirable property values, we use a guidance term added to the score during reverse diffusion. At timestep t the conditional guided score is:

$$s_{\text{guided}}(y_t, t) = s_\theta(y_t, t \mid h_p, h_c, g) + \sum_{k=1}^K \lambda_k(t) \nabla_{y_t} \log p_{\phi_k}(s_k^* \mid y_t), \quad (6)$$

where s_k^* denotes target property preferences (e.g., high BBB probability, high LNP-encapsulation score), and $\lambda_k(t)$ are time-dependent guidance weights that modulate influence at different noise scales. This is analogous to classifier-guidance and score-based guidance strategies but extended to multiple, possibly competing objectives and implemented within an SE(3)-equivariant architecture [3].

We integrate a differentiable retrosynthesis/scoring module $R_\omega(y)$ that outputs a continuous synthetic-feasibility reward and, when possible, a differentiable surrogate for step-count or route-likelihood. The retrosynthesis term enters the guidance sum above (as one of the k tasks) and encourages generation of synthetically plausible scaffolds. This design is informed by prior work that couples generation with retrosynthetic feasibility scoring [4, 5].

Rationale: Making retrosynthesis scoring differentiable allows the sampler to favor chemically reachable structures during the generative trajectory, reducing the post-hoc rejection rate and improving the chance that top-ranked designs have plausible synthetic routes.

Multi-objective optimization: uncertainty-aware scalarization and Pareto considerations Because objectives (CNS activity, BBB, LNP compatibility, synthetic-feasibility) can conflict, we adopt a hybrid approach combining adaptive scalarization and Pareto-aware selection. At generation time we construct an adaptive scalar reward

$$R_{\text{agg}}(y) = \sum_{k=1}^K w_k \tilde{r}_k(y), \quad (7)$$

where $\tilde{r}_k(y)$ are normalized task-specific rewards and weights w_k are adapted online using predictor uncertainties $\hat{\sigma}_k$:

$$w_k \propto \frac{1}{\hat{\sigma}_k + \epsilon}, \quad (8)$$

so that tasks with lower epistemic uncertainty exert more influence. In parallel, we maintain a candidate archive and compute Pareto-frontier sets with respect to key task axes; samples on or near the Pareto front are prioritized for downstream evaluation and acquisition. Using both scalarization (to guide diffusion trajectories) and explicit Pareto selection (to preserve trade-off diversity) balances focused optimization with exploration of diverse trade-offs.

Rationale: Uncertainty-weighted scalarization reduces the risk of over-optimizing on poorly calibrated predictors, while Pareto selection preserves multiple clinically relevant trade-offs rather than collapsing to a single weighted objective.

Active learning loop and acquisition To mitigate data scarcity for CNS-specific labels, we embed the generator and predictors in an active-learning loop. Let $\mathcal{D}_L$ be labeled data and $\mathcal{D}_U$ unlabeled candidate pool. After generating a batch of candidates $\{y^{(i)}\}$, we compute acquisition scores

$$\alpha(y) = \sum_{k=1}^K \beta_k \text{acq}_k(y), \quad (9)$$

where $\text{acq}_k(y)$ is an acquisition metric (e.g., expected improvement, BALD, or uncertainty sampling) for task k , and β_k are task-prior weights. Selected candidates are labeled by high-fidelity in silico surrogates or targeted experiments, appended to $\mathcal{D}_L$, and predictors ϕ_k are updated. The generator θ and encoder ψ are then fine-tuned or re-calibrated on updated conditioning distributions.

Rationale: Iterative acquisition focuses scarce experimental resources on candidates that maximally reduce predictor uncertainty and improve generator performance. Collectively, these components form a replicable pipeline that produces 3D, pharmacophore-aligned molecular proposals tailored for CNS activity, BBB penetration propensity, and LNP deliverability while explicitly accounting for synthetic feasibility.

Algorithmic schema The overall pipeline is summarized in Algorithm 1. The generation loop alternates between conditioned SE(3)-equivariant sampling with multi-task guidance, differentiable retrosynthesis scoring, Pareto-based selection, acquisition-driven labeling, and predictor updates.

Implementation considerations and summary Our method integrates pharmacophore-conditioned SE(3)-equivariant diffusion to respect 3D interaction constraints [2, 3], differentiable retrosynthesis scoring to favor synthetically accessible chemistries [4, 5], and uncertainty-aware multi-objective guidance to reconcile competing CNS delivery and activity objectives. The active-learning loop focuses labeling effort where it most reduces uncertainty and improves generator performance. Collectively, these components form a replicable pipeline that produces 3D, pharmacophore-aligned molecular proposals tailored for CNS activity, BBB penetration propensity, and LNP deliverability while explicitly accounting for synthetic feasibility.

4 Experimental Setup

This section describes the datasets, preprocessing, model training, evaluation metrics, baselines, ablation studies, and the stepwise experimental plan that will be used to validate NeuroPharmDiff-MultiBio. The objective is to provide sufficient detail for reproducibility and to justify design choices that follow from the Method.

4.1 Datasets and selection rationale

Primary molecular datasets. We will assemble a CNS-focused dataset by integrating (i) public BBB permeability and brain-penetrant drug annotations, (ii) bioactivity records for selected CNS targets from curated sources (e.g., ChEMBL subsets), and (iii) synthetic-feasibility and retrosynthesis-route corpora derived from public reaction/retrosynthesis

Algorithm 1 NeuroPharmDiff-MultiBio: generation + active learning loop

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Initialize generator  $\theta$ , pharmacophore encoder  $\psi$ , predictors  $\{\phi_k\}$ , retrosynthesis scorer  $\omega$ 
for each active-learning iteration do
  Sample pharmacophore hypotheses  $\{p_i\}$  and optional pockets  $\{g_i\}$ 
  for each  $p_i$  do
    Compute  $h_p \leftarrow \mathcal{E}_\psi(p_i)$ 
    Sample  $y_T \sim$  prior noise
    for  $t = T$  to 1 do
      Compute base score  $s_\theta(y_t, t \mid h_p, h_c, g)$ 
      Compute property gradients  $\nabla_{y_t} \log p_{\phi_k}(s_k^* \mid y_t)$  for all  $k$ 
      Form guided score  $s_{\text{guided}}$  and step to  $y_{t-1}$ 
    end for
    Evaluate differentiable retrosynthesis score  $R_\omega(y_0)$ 
    Compute multi-objective rewards and acquisition score  $\alpha(y_0)$ 
  end for
  Select top candidates by Pareto or  $\alpha$  and query labels (in silico/in vitro)
  Update predictors  $\{\phi_k\}$  using new labels
  Optionally fine-tune  $\theta, \psi, \omega$  on revised objectives
end for

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datasets. Dataset construction emphasizes molecules with experimentally measured BBB or CNS exposure whenever available; where direct measurements are absent, high-confidence annotations (e.g., labeled CNS drugs or validated literature reports) are used conservatively. The assembled training corpus will therefore contain a mix of experimentally labeled examples and larger pools of unlabeled or weakly labeled molecules for generator pretraining and active labeling.

LNP-compatibility annotations. LNP encapsulation and formulation-relevant labels are sparse in public repositories. We will therefore curate an LNP-feasibility corpus combining (i) public reports of small-molecule LNP encapsulation success/failure, (ii) physicochemical property heuristics known to influence encapsulation (lipophilicity, pKa proxies), and (iii) proxy labels derived from formulation-focused studies. These form an initial labeled set for the LNP predictor and a larger unlabeled pool for active-label acquisition.

Train / validation / test splits. For each labeled task (BBB, target activity, LNP), we will partition data into disjoint train/validation/test splits with scaffold-based separation to minimize scaffold leakage. Specifically, molecules are grouped by Bemis–Murcko scaffold and splits are constructed so that no scaffold appears in more than one split. Where label counts are low (per-task < a few thousand), we reserve at least 20% of labeled molecules for validation and 10% for a held-out test set; for very small tasks we use nested cross-validation as a robustness check.

4.2 Preprocessing and 3D representations

Standardization and sanitization. All molecules are standardized (aromaticity normalization, salt stripping, tautomer normalization) using established chemoinformatics toolkits. Atom typing follows a fixed schema used across predictors and the generator.

Conformer generation. For each molecule we generate up to $M = 8$ low-energy 3D conformers using an established distance-geometry / force-field workflow, and retain the lowest-energy conformer for primary training while preserving alternative conformers for uncertainty estimation and augmentation during predictor training.

Pharmacophore encoding. Pharmacophore hypotheses are encoded into continuous vectors $h_p \in \mathbb{R}^{d_p}$ with $d_p = 128$ by combining type, relative distances, and angular constraints; this encoder is permutation-invariant over pharmacophore points and yields multi-scale spatial features that are concatenated to node-level and global conditioning channels of the SE(3)-equivariant backbone.

4.3 Model architecture and implementation details

Generator. The generative backbone is an SE(3)-equivariant diffusion model that jointly models node attributes, pairwise bond logits, and Cartesian coordinates. The score network uses stacked SE(3)-equivariant message-passing blocks with spherical-harmonic inspired geometric features to preserve equivariance and to support coordinate gradients required for guidance; this design follows SE(3)-aware architectures used in structure-aware diffusion work [3, 3].

Property predictors. Multi-task predictors f_k share a common 3D-graph encoder backbone and have task-specific heads that output both a score $\hat{s}_k$ and an uncertainty estimate $\hat{\sigma}_k$. Predictors are trained with log-likelihood or proper scoring rules to encourage calibrated uncertainties; calibrated uncertainties are used to set guidance weights and to prioritize acquisition in active learning.

Differentiable retrosynthesis module. The retrosynthesis scorer R_ω is implemented as a differentiable surrogate trained to predict route-likelihood and approximate synthetic step counts from molecule encodings; development follows approaches that couple generation with retrosynthetic feasibility scoring [4, 5]. The scorer provides a continuous reward and, where available, a proposed route for downstream manual inspection.

Guidance and multi-objective optimizer. During reverse diffusion we compute guided scores by adding gradient-based nudges from each differentiable predictor; guidance scheduling and uncertainty-weighted scalarization follow the formulation in the Method (weights adapt proportional to $1/(\hat{\sigma}_k + \epsilon)$). SE(3)-equivariant implementation ensures gradients respect geometric equivariance and are computed with autodiff through the predictor stack.

Software and hardware. Models will be implemented in PyTorch with an SE(3)-equivariance library for geometric layers and trained on a multi-node A100 cluster (target configuration: 8x A100 GPUs per training run). Docking and physics-based rescoring use established open-source tools wrapped into reproducible pipelines.

4.4 Training protocol and hyperparameters

Pretraining and fine-tuning schedule. The generator is pretrained on a broad molecule corpus (large, unlabeled) to learn general 3D chemical priors, then fine-tuned on the CNS-focused corpus with pharmacophore and property conditioning. Predictors are trained jointly as a multi-task model on labeled examples; they are periodically fine-tuned during active-learning iterations with newly acquired labels.

Optimization and baseline hyperparameters. Default hyperparameters used across experiments are:

- Optimizer: AdamW with weight decay $1\text{e}-2$.
- Learning rate (generator and predictors): $1\text{e}-4$, with alternative runs at $3\text{e}-4$ for sensitivity analysis.
- Batch size: 16 for generator training; effective per-GPU batches adjusted with gradient accumulation as needed.
- Number of diffusion timesteps: $T = 1000$.
- Conditioning dimension: $d_p = 128$ (pharmacophore), $d_c = 128$ (auxiliary).
- Epoch budget for fine-tuning: up to 100 epochs with early stopping on validation composites.

Guidance scheduling. Time-dependent guidance weights $\lambda_k(t)$ are annealed to focus property gradients at intermediate-to-late noise scales; we will sweep three canonical schedules (early-heavy, late-heavy, uniform) and report sensitivity in ablation studies.

4.5 Baselines and ablation studies

Baselines. We will compare against:

1. An RFDiffusion-style 3D diffusion baseline without pharmacophore conditioning (generator-only baseline) [2].
2. An SE(3)-equivariant diffusion conditioned on simple BBB-predictor scores but without LNP or retrosynthesis conditioning [3].
3. Rule-based pharmacophore-guided generation (non-diffusion heuristic baseline).
4. An unconditional backbone diffusion model (no conditioning).

Ablations. We will perform targeted ablations to measure contribution of individual components:

- Remove differentiable retrosynthesis scoring to assess impact on synthetic-feasibility rates.
- Replace uncertainty-weighted scalarization with fixed weighting to evaluate robustness.
- Disable LNP predictor guidance to quantify influence on formulation compatibility.
- Vary pharmacophore encoding dimensionality ($d_p \in \{64, 128, 256\}$).

4.6 Evaluation metrics and protocols

Property prediction metrics. For predictors we report ROC-AUC (classification tasks), RMSE / R^2 for regression tasks, calibration metrics (expected calibration error), and per-task uncertainty calibration plots.

Generative metrics. For generated sets we compute:

- Novelty: fraction of generated molecules not present in training data.
- Validity: fraction of chemically valid molecules.
- Uniqueness: fraction of unique molecules in the generated set.
- Synthetic-feasibility rate: fraction of molecules with retrosynthesis score above a threshold or with plausible step-count $\leq$ predefined cutoff.
- Predicted BBB-pass rate: fraction of generated molecules with predicted BBB probability above target threshold.
- LNP-compatibility score distribution: summary statistics of predicted encapsulation propensity.
- Diversity: internal chemical diversity measured by pairwise fingerprint distance statistics.

Multi-objective evaluation. We analyze trade-offs using Pareto-frontier constructions across axes (predicted CNS activity, BBB probability, LNP score, synthetic feasibility). For sets of generated samples we compute hypervolume dominated by the Pareto front as an aggregate indicator of multi-objective progress.

Downstream structural validation. Top candidates will be evaluated with docking and structure-aware rescoring to assess putative binding modes and shape complementarity; the choice of docking protocol follows prior structure-conditioned diffusion studies [3]. For a subset, molecular dynamics based stability checks will be performed when computational budget permits.

Experimental validation. Where feasible, a small number of top-ranked molecules will be prioritized for synthesis planning and limited in vitro assays (permeability proxies, target binding) as part of the active-learning label acquisition. The decision to forward a candidate to experimental validation will depend on combined predicted performance, retrosynthesis route quality, and cost estimates.

4.7 Reproducibility and statistical rigor

Random seeds and runs. All experiments report results aggregated over multiple independent runs (minimum of three seeds) and provide standard errors. Seeds for data splits, weight initialization, and sampling are fixed and reported for each experiment.

Hyperparameter reporting. We will publish the full hyperparameter sweeps and the configuration files used for the reported runs. For each reported model, we list the final trained checkpoint hash, training duration, and hardware used.

Code and data release. To enable replication we will release code, training scripts, dataset assembly pipelines (subject to licensing), and model checkpoints, together with a README that reproduces the main tables and figures.

4.8 Experimental plan and timeline

The experimental campaign is organized into three phases.

Phase I – Baselines and core-component validation (Months 0–4).

- Assemble datasets, set up preprocessing and conformer pipelines.
- Pretrain generator on large unlabeled corpus; train initial multi-task predictors.
- Run baseline comparisons: unconditional diffusion, RFDiffusion-style baseline, and SE(3) diffusion with simple BBB guidance.
- Evaluate core metrics (validity, novelty, predictor performance).

Phase II – Integrated NeuroPharmDiff training and ablations (Months 4–9).

- Train full NeuroPharmDiff with pharmacophore conditioning, differentiable retrosynthesis, and LNP predictor.
- Conduct ablation studies and guidance-schedule sweeps.

- Construct Pareto front analyses and hypervolume comparisons versus baselines.
- Select top candidate pools for active acquisition.

Phase III – Active learning loop and limited experimental validation (Months 9–15).

- Run iterative active-learning cycles: generate candidates, acquire in silico labels or targeted in vitro labels, update predictors, and fine-tune the generator.
- For highest-confidence candidates with high retrosynthesis scores, perform synthesis planning and, resources permitting, small-scale synthesis and permeability/target-binding assays.
- Report how active labeling improves predictor calibration, generator yield of high-quality candidates, and Pareto frontier expansion.

4.9 Resource estimates and stopping criteria

We plan to budget compute for several full training cycles on an 8xA100 configuration; each full fine-tuning run is expected to take on the order of days to a few weeks depending on dataset size and augmentation. Active-learning cycles will be limited by labeling throughput: we will typically acquire tens to low hundreds of new high-fidelity labels per cycle and stop when marginal utility of new labels (measured by reduction in predictor uncertainty and Pareto-front improvement) falls below a preset threshold.

4.10 Risk mitigation and contingency plans

Data sparsity for LNP and BBB labels is a primary risk. To mitigate this we will:

- Use calibrated uncertainty and conservative weighting to avoid overfitting to weak labels.
- Leverage transfer learning from larger molecular property datasets during predictor pretraining.
- Use in silico surrogates and orthogonal physics-based checks (docking, simple permeability models) to triage candidates before wet-lab validation [3].

4.11 Summary

The experimental setup combines scaffold-split evaluation, calibrated multi-task predictors, SE(3)-equivariant generative modeling, differentiable retrosynthesis scoring, and a principled active-learning acquisition loop. Baselines and ablations are designed to isolate the contribution of pharmacophore conditioning, LNP guidance, and retrosynthesis integration; multi-objective metrics and Pareto analyses will quantify progress toward simultaneous CNS activity, BBB penetration propensity, LNP deliverability, and synthetic feasibility. The experimental protocol emphasizes reproducibility (fixed seeds, multiple runs), transparency (release of code and configs), and pragmatism (prioritizing candidates amenable to synthesis and validation). References to prior work motivate architectural and evaluation choices [2, 3, 4, 5, 1].

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